



Delivery Delay Prediction in Logistics Systems

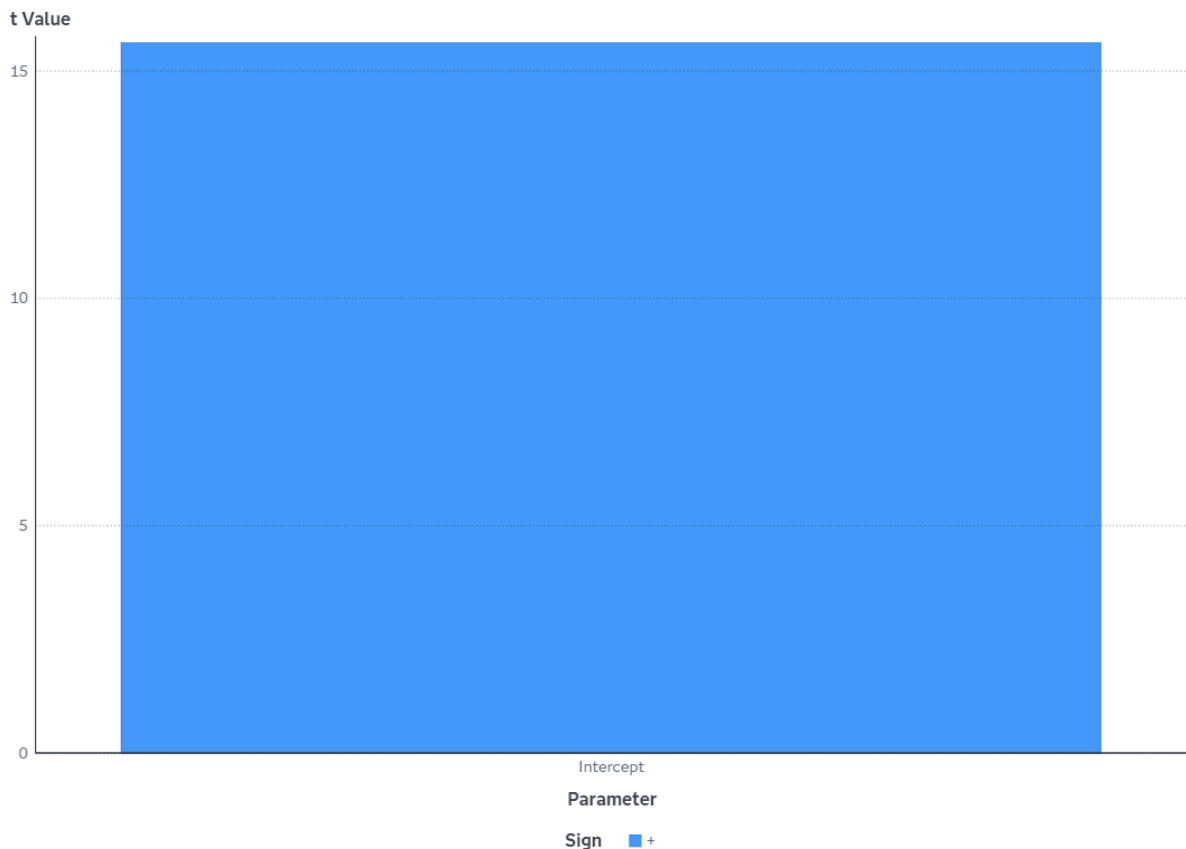
"Logistic Regression" Results

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Contents

t Values by Parameter	3
Parameter Estimates	4
Selection Summary	5
Regression Fit Statistics	6
Score Outputs	7
Cumulative Lift	9
Lift	11
Gain	13
Captured Response Percentage	15
Cumulative Captured Response Percentage	16
Response Percentage	18
Cumulative Response Percentage	19
ROC	21
Accuracy	23
F1 Score	24
Fit Statistics	26
Percentage Plot	27
Count Plot	28
Table	29
Properties	31
Output	33

t Values by Parameter



This plot displays the absolute value of the t value for each parameter estimate in the logistic regression model. Larger values indicate more significant parameters. The bar that represents the parameter is colored by the sign of the estimate. Bars that are colored as positive (+) correspond to a positive parameter estimate, which indicates an increase in the predicted probability of the event as the parameter value increases. Bars that are colored as negative (-) correspond to a negative parameter estimate, which indicates a decrease in the predicted probability of the event as the parameter value increases. The most significant parameter is Intercept with a t value of 15.62.

Parameter Estimates

Effect	Parameter	t Value	Sign
Intercept	Intercept	15.6198	+

Estimate	Absolute Estimate	Standard Error	Chi-Square
0.3920	0.3920	0.0251	243.9778

Pr > Chi-Square	Degrees of Freedom
0.0000	1

Selection Summary

Step	Effect Entered	Effect Removed	Number of Effects
0	Intercept		1
1	Discount_offered		2
2		Discount_offered	1

SBC	Optimal SBC
8,908.2677	0
7,900.5241	0
7,894.2982	1

Regression Fit Statistics

Statistic	Description	Training	Validation
M2LL	-2 Log Likelihood	8,899.4730	4,450.6447
AIC	AIC (smaller is better)	8,901.4730	4,452.6447
AICC	AICC (smaller is better)	8,901.4736	4,452.6459
SBC	SBC (smaller is better)	8,908.2677	4,458.7464
ASE	Average Square Error	0.2406	0.2407

Testing
1,483.8096
1,485.8096
1,485.8132
1,490.8126
0.2407

Score Outputs

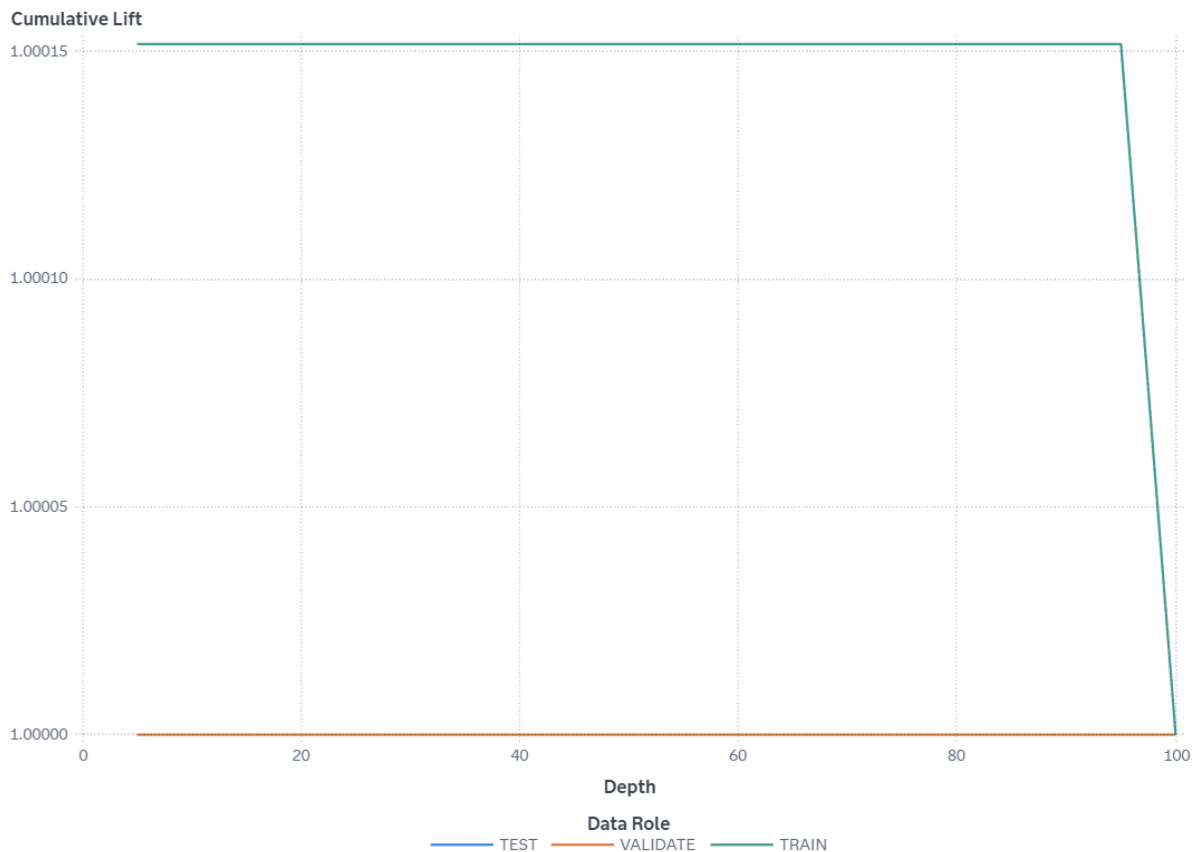
Name	Role	Type	Variable Type
EM_CLASSIFICATION	CLASSIFICATION	C	char
EM_EVENTPROBABILITY	PREDICT	N	double
EM_PROBABILITY	PREDICT	N	double
I_Reached_on_Time_Y_N	CLASSIFICATION	C	char
P_Reached_on_Time_Y_N0	PREDICT	N	double
P_Reached_on_Time_Y_N1	PREDICT	N	double

Variable Label	Variable Format	Variable Length	Creator
Predicted for Reached.on.Time_Y.N		12	logisticreg
Probability for Reached.on.Time_Y.N=1		8	logisticreg
Probability of Classification		8	logisticreg
Into: Reached.on.Time_Y.N		12	logisticreg
Predicted: Reached.on.Time_Y.N=0		8	logisticreg
Predicted: Reached.on.Time_Y.N=1		8	logisticreg

Function	Creator GUID
CLASSIFICATION	85c49787-4dfe-46f2-9524-94e3291045ec

Function	Creator GUID
PREDICT	85c49787-4dfe-46f2-9524-94e3291045ec
PREDICT	85c49787-4dfe-46f2-9524-94e3291045ec
CLASSIFICATION	85c49787-4dfe-46f2-9524-94e3291045ec
PREDICT	85c49787-4dfe-46f2-9524-94e3291045ec
PREDICT	85c49787-4dfe-46f2-9524-94e3291045ec

Cumulative Lift



The VALIDATE partition has a Cumulative Lift of 1 in the 10% quantile (depth of 10) meaning there are 1 times fewer events in the first two quantiles than expected by random (10% of the total number of events). Because this value is less than 1, it is worse to use your model to identify responders than no model, based on the selected partition.

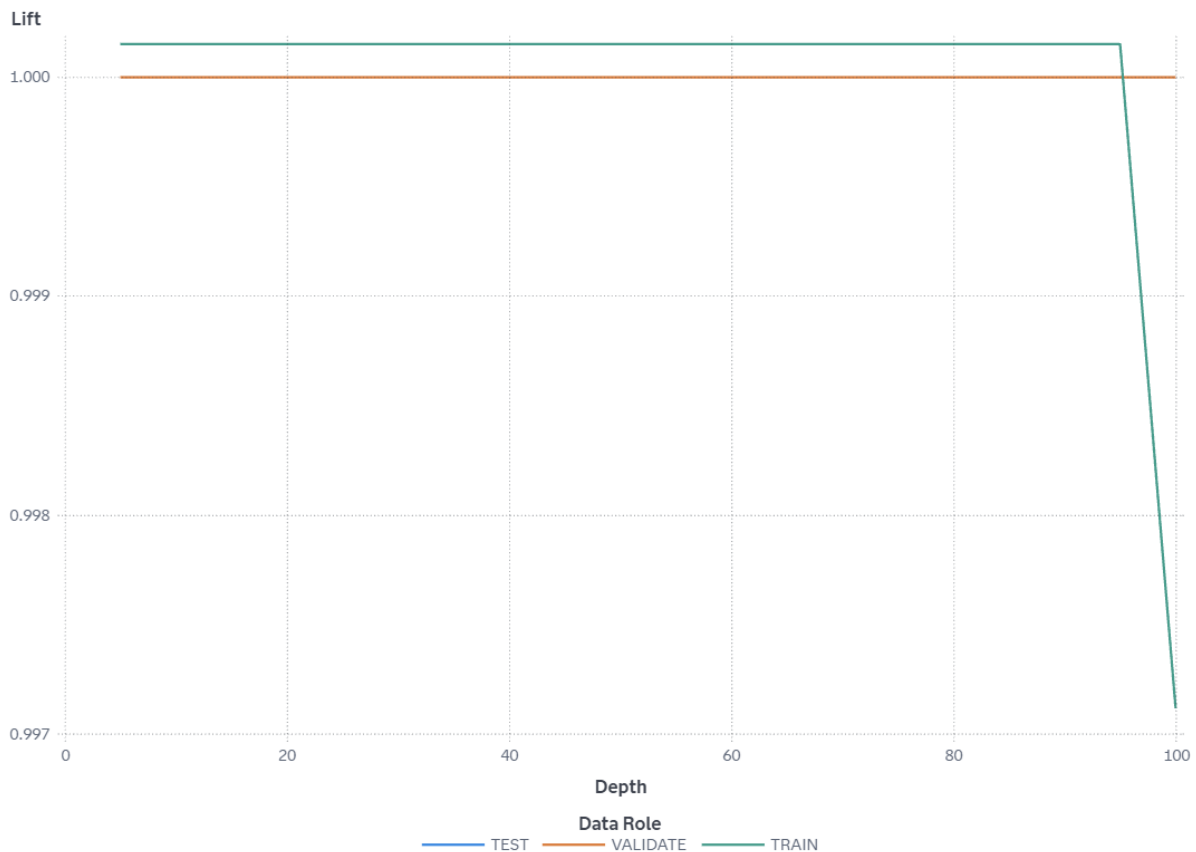
The TRAIN partition has a Cumulative Lift of 1 in the 10% quantile (depth of 10) meaning there are 1 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Cumulative Lift of 1 in the 10% quantile (depth of 10) meaning there are 1 times fewer events in the first two quantiles than expected by random (10% of the total number of events). Because this value is less than 1, it is worse to use your model to identify responders than no model, based on the selected partition.

Cumulative lift is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Reached_on_Time_Y_N1`, which represents the predicted probability of the event "1" for the target

Reached.on.Time_Y.N. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative lift for a particular quantile is the ratio of the number of events across all quantiles up to and including the current quantile to the number of events that would be there at random, or equivalently, the ratio of the cumulative response percentage to the baseline response percentage. The cumulative lift at depth 10 includes the top 10% of the data, which is the first 2 quantiles, which would have 10% of the events at random. Thus, cumulative lift measures how much more likely it is to observe an event in the quantiles than by selecting observations at random.

Lift



The VALIDATE partition has a Lift of 1 in the 5% quantile (depth of 5) meaning there are 1 times fewer events in that quantile than expected by random (5% of the total number of events). Because this value is less than 1, it is worse to use your model to identify responders than no model, based on the selected partition.

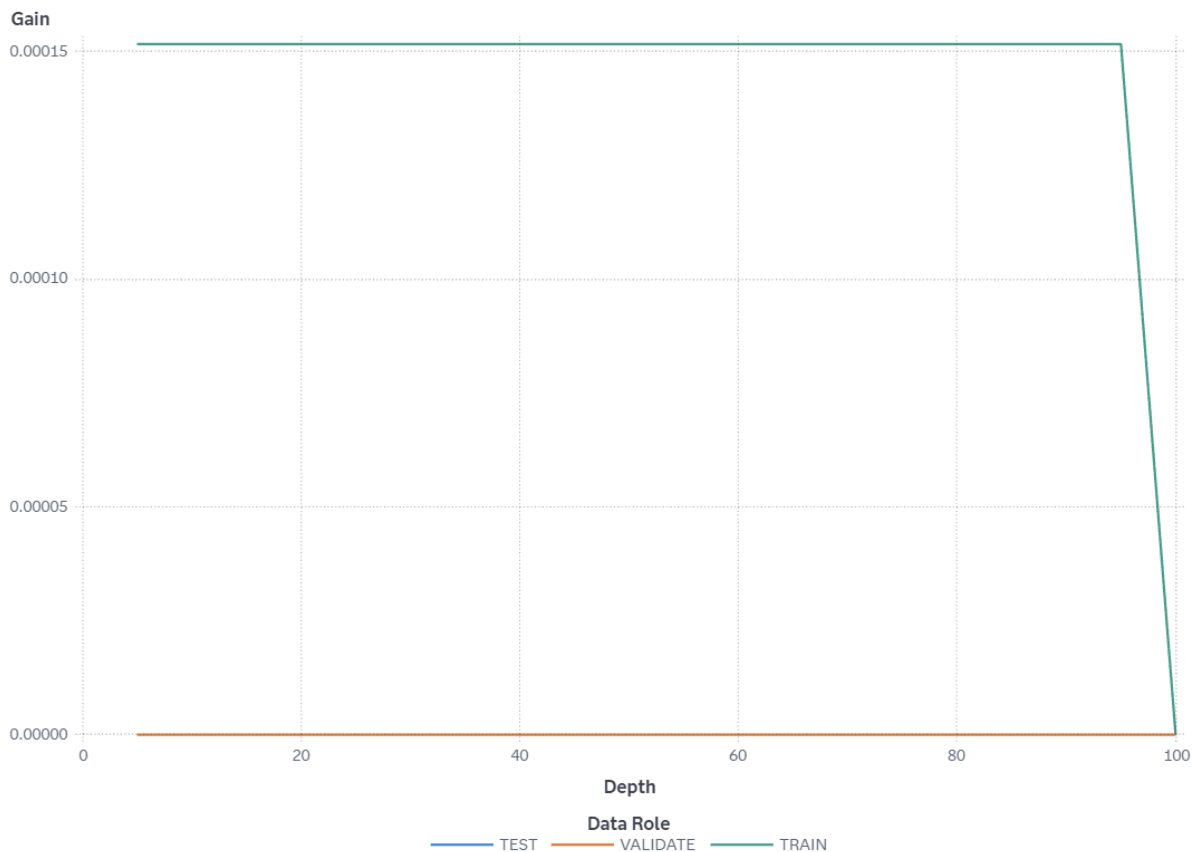
The TRAIN partition has a Lift of 1 in the 5% quantile (depth of 5) meaning there are 1 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Lift of 1 in the 5% quantile (depth of 5) meaning there are 1 times fewer events in that quantile than expected by random (5% of the total number of events). Because this value is less than 1, it is worse to use your model to identify responders than no model, based on the selected partition.

Lift is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Reached_on_Time_Y_N1`, which represents the predicted probability of the event "1" for the target `Reached.on.Time_Y.N`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Lift is the ratio of the number of events in that quantile to the number of events that would be there at random, or

equivalently, the ratio of the response percentage to the baseline response percentage. With 20 quantiles, it is expected that 5% of the events occur in each quantile. Thus, Lift measures how much more likely it is to observe an event in each quantile than by selecting observations at random.

Gain



The VALIDATE partition has a Gain of 0 at the 10% quantile (depth of 10). Because this value is less than 0, it is worse to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

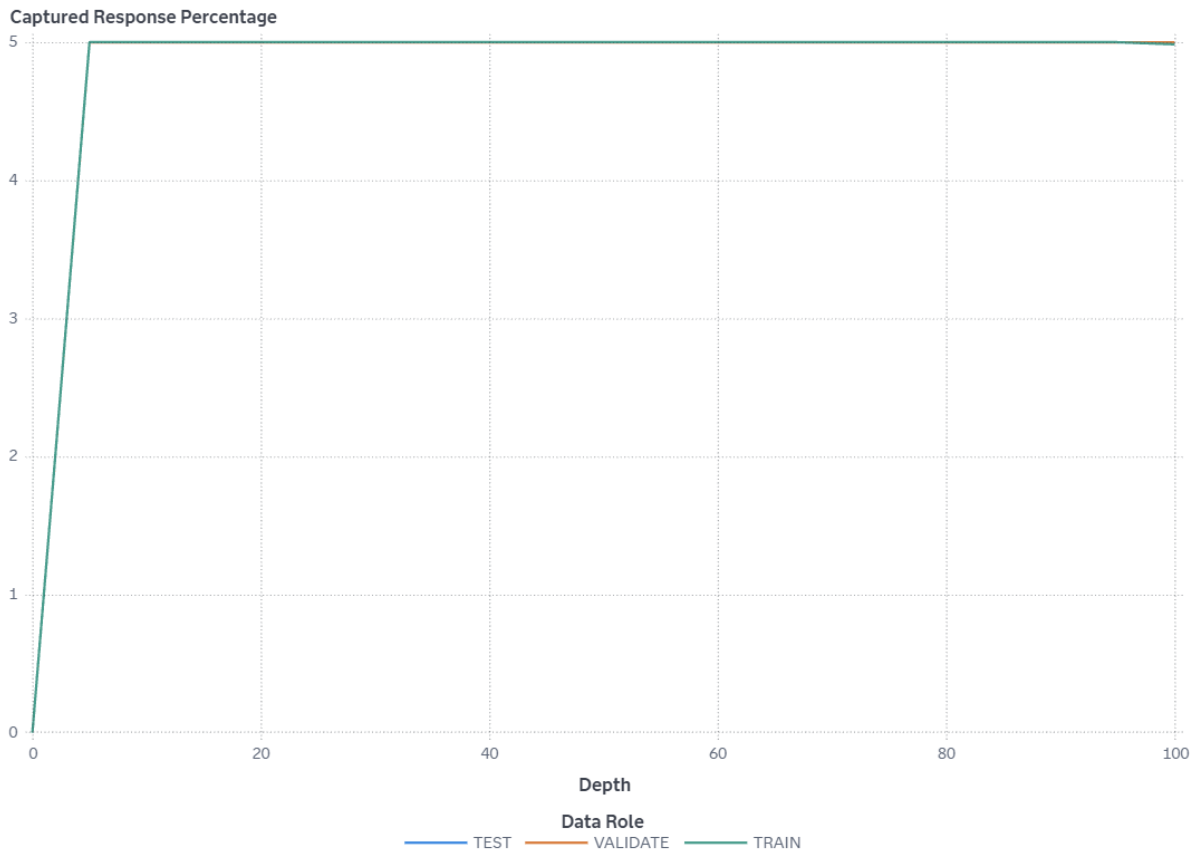
The TRAIN partition has a Gain of 0 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

The TEST partition has a Gain of 0 at the 10% quantile (depth of 10). Because this value is greater than 0, it is better to use your model to identify responders than no model, based on the selected partition. The best possible value of Gain for this partition at depth 10 is 0.68.

Gain is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Reached_on_Time_Y_N1`, which represents the predicted probability of the event "1" for the target `Reached.on.Time_Y.N`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Gain is a cumulative measure for the quantiles up to and including the current one and is calculated as (number of events

in the quantiles) / (number of events expected by random) - 1. With 20 quantiles, it is expected that 5% of the events occur in each quantile. Note that the value of Gain is the same as the value of Cumulative Lift - 1. If the value of Gain is greater than 0, then your model is better at identifying events than using no model.

Captured Response Percentage



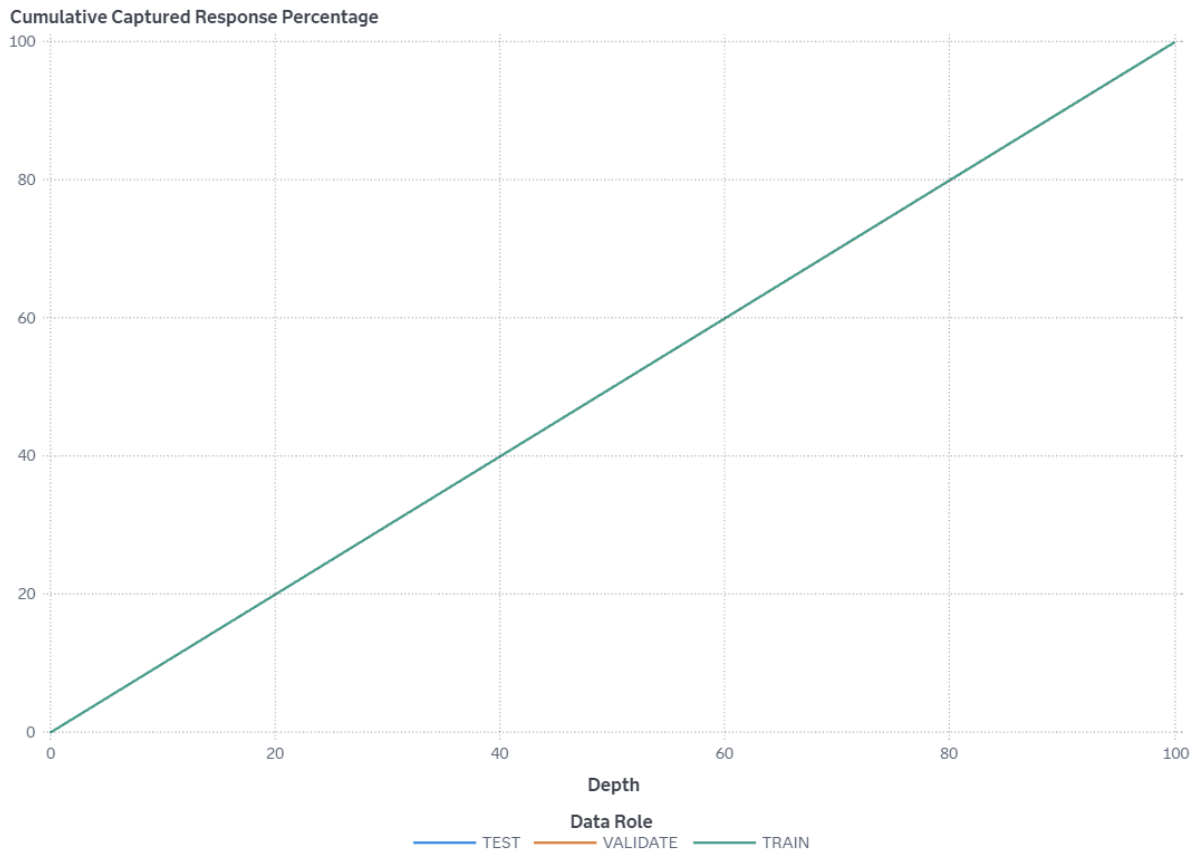
At the 5% quantile (depth of 5), the VALIDATE partition has a Captured response percentage of 5 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

At the 5% quantile (depth of 5), the TRAIN partition has a Captured response percentage of 5 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

At the 5% quantile (depth of 5), the TEST partition has a Captured response percentage of 5 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 8.38.

Captured response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Reached_on_Time_Y_N1`, which represents the predicted probability of the event "1" for the target `Reached.on.Time_Y.N`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Captured response percentage is the percentage of the total number of events that are in that quantile. With no model, it is expected that 5% of the events are in each quantile.

Cumulative Captured Response Percentage



In the top 10% of the data (depth 10), the VALIDATE partition has a Cumulative captured response percentage of 10 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.76.

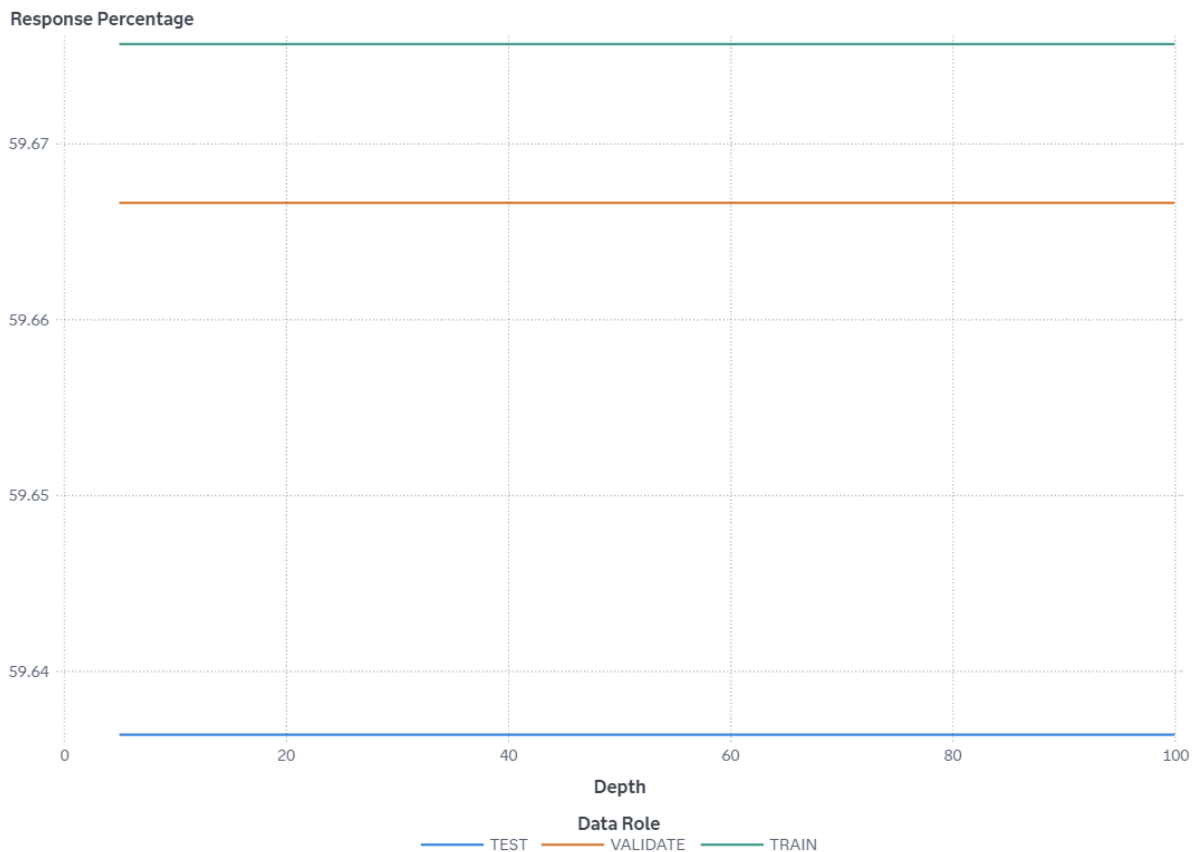
In the top 10% of the data (depth 10), the TRAIN partition has a Cumulative captured response percentage of 10 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.76.

In the top 10% of the data (depth 10), the TEST partition has a Cumulative captured response percentage of 10 (compared to the expected value of 10 for no model). The best possible value of Cumulative captured response percentage for this partition at depth 10 is 16.77.

Cumulative captured response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Reached_on_Time_Y_N1`, which represents the predicted probability of the event "1" for the target `Reached.on.Time_Y.N`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative captured response percentage for a particular quantile is

the percentage of the total number of events that are in the quantiles up to and including the current quantile. With no model, it is expected that 5% of the events are in each quantile, so the cumulative captured response percentage at depth 10 would be 10%.

Response Percentage



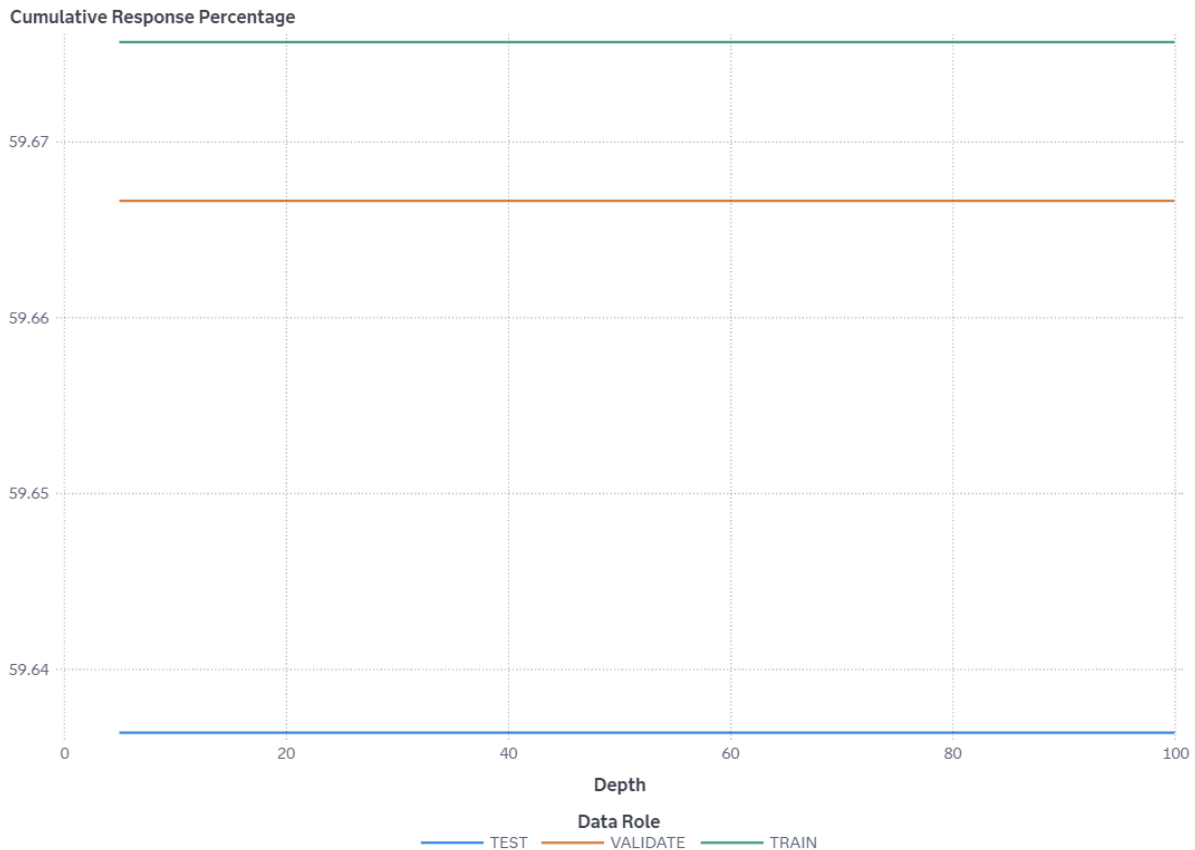
At the 5% quantile (depth of 5), the VALIDATE partition has a Response percentage of 59.7. The best possible value of Response percentage for this partition at depth 5 is 100.

At the 5% quantile (depth of 5), the TRAIN partition has a Response percentage of 59.7. The best possible value of Response percentage for this partition at depth 5 is 100.

At the 5% quantile (depth of 5), the TEST partition has a Response percentage of 59.6. The best possible value of Response percentage for this partition at depth 5 is 100.

Response percentage is calculated by sorting each partition in descending order by the predicted probability of the target event `P_Reached_on_Time_Y_N1`, which represents the predicted probability of the event "1" for the target `Reached.on.Time_Y.N`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Response percentage is the percentage of observations that are events in that quantile. With no model, it is expected that the response percentage is constant across quantiles, $100 \times \text{overall-event-rate}$. This is also called the baseline response percentage.

Cumulative Response Percentage



In the top 10% of the data (depth 10), the VALIDATE partition has a Cumulative response percentage of 59.7. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

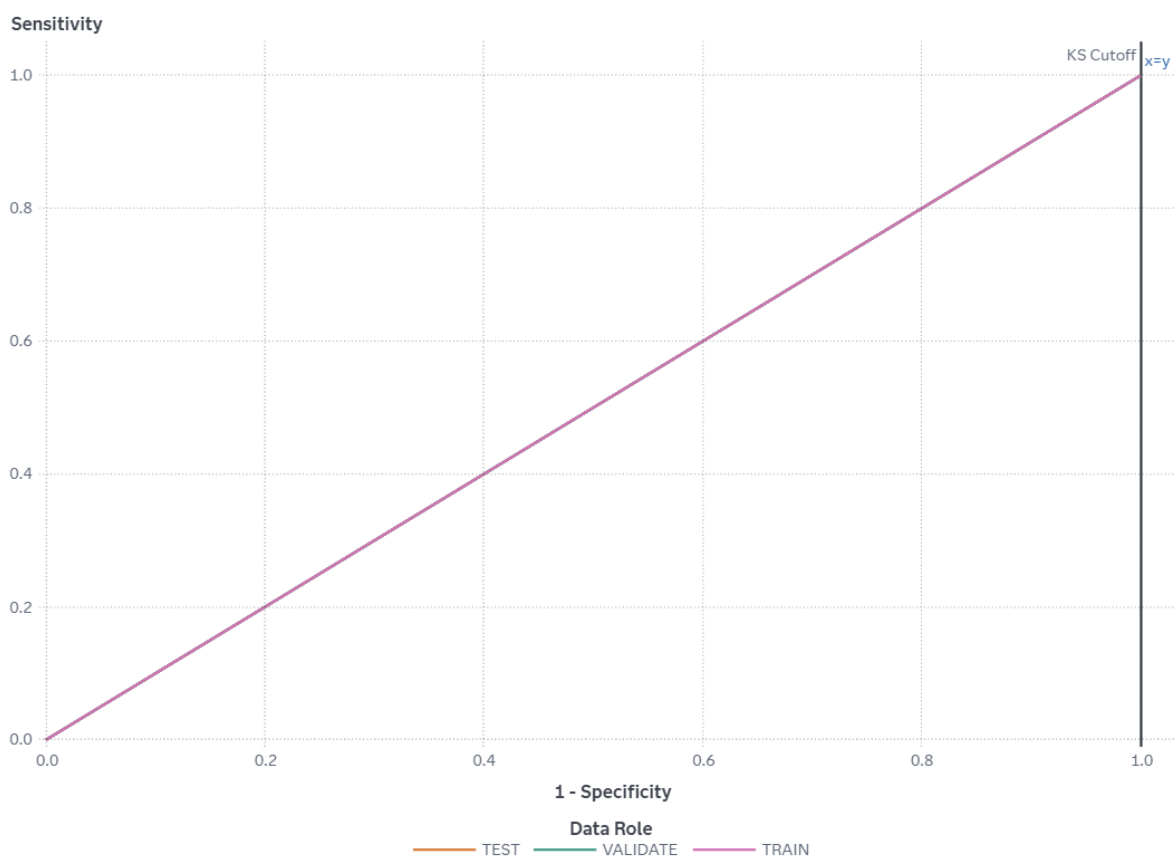
In the top 10% of the data (depth 10), the TRAIN partition has a Cumulative response percentage of 59.7. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

In the top 10% of the data (depth 10), the TEST partition has a Cumulative response percentage of 59.6. The best possible value of Cumulative response percentage for this partition at depth 10 is 100.

Cumulative response percentage is calculated by sorting in descending order each partition of the data by the predicted probability of the target event `P_Reached_on_Time_Y_N1`, which represents the predicted probability of the event "1" for the target `Reached.on.Time_Y.N`. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative response percentage for a particular quantile is the percentage of observations that are events in the quantiles up to and including the current quantile. With no model, it is expected that the response percentage is constant across quantiles, $100 \times \text{overall-event-rate}$. This is also called the baseline

response percentage.

ROC



The ROC curve is a plot of sensitivity (the true positive rate) against 1-specificity (the false positive rate), which are both measures of classification based on the confusion matrix. These measures are calculated at various cutoff values. To help identify the best cutoff to use when scoring your data, the KS Cutoff reference line is drawn at the value of 1-specificity where the greatest difference between sensitivity and 1-specificity is observed for the VALIDATE partition. The KS Cutoff line is drawn at the cutoff value 0, where the 1-specificity value is 1 and the sensitivity value is 1.

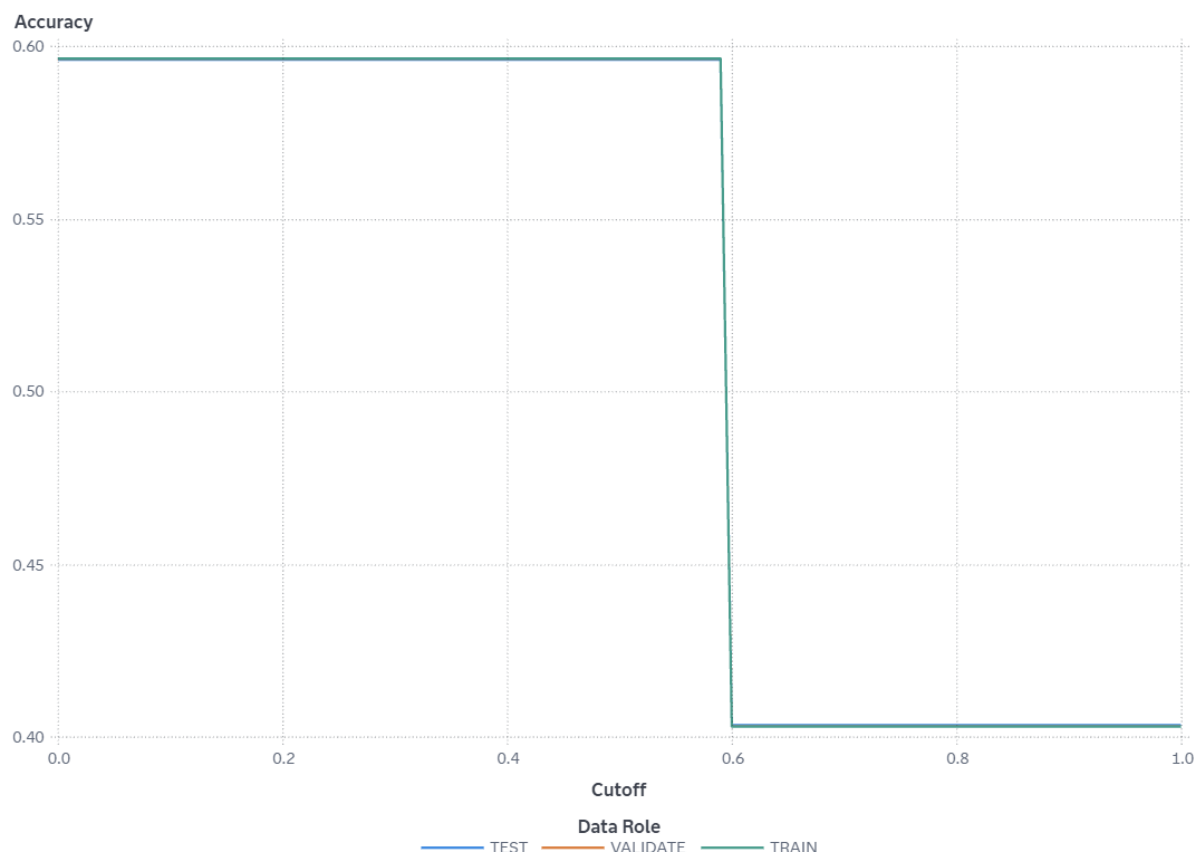
Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether `P_Reached_on_Time_Y_N1`, which is the predicted probability of the event "1" for the target `Reached.on.Time_Y.N`, is greater than or equal to the cutoff value. When `P_Reached_on_Time_Y_N1` is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event.

The confusion matrix for each cutoff value contains four cells that display the true positives for events that are correctly classified (TP), false positives for non-events that are classified as events (FP), false negatives for events that are classified as non-events (FN), and true negatives for non-events that are classified as non-events (TN). True negatives include non-event classifications that specify a different non-event. Sensitivity is calculated as $TP / (TP + FN)$. Specificity, the true negative rate,

is calculated as $TN / (TN + FP)$, so $1 - \text{specificity}$ is $FP / (TN + FP)$. The values of sensitivity and $1 - \text{specificity}$ are plotted at each cutoff value.

A ROC curve that rapidly approaches the upper-left corner of the graph, where the difference between sensitivity and $1 - \text{specificity}$ is the greatest, indicates a more accurate model. A diagonal line where sensitivity = $1 - \text{specificity}$ indicates a random model.

Accuracy



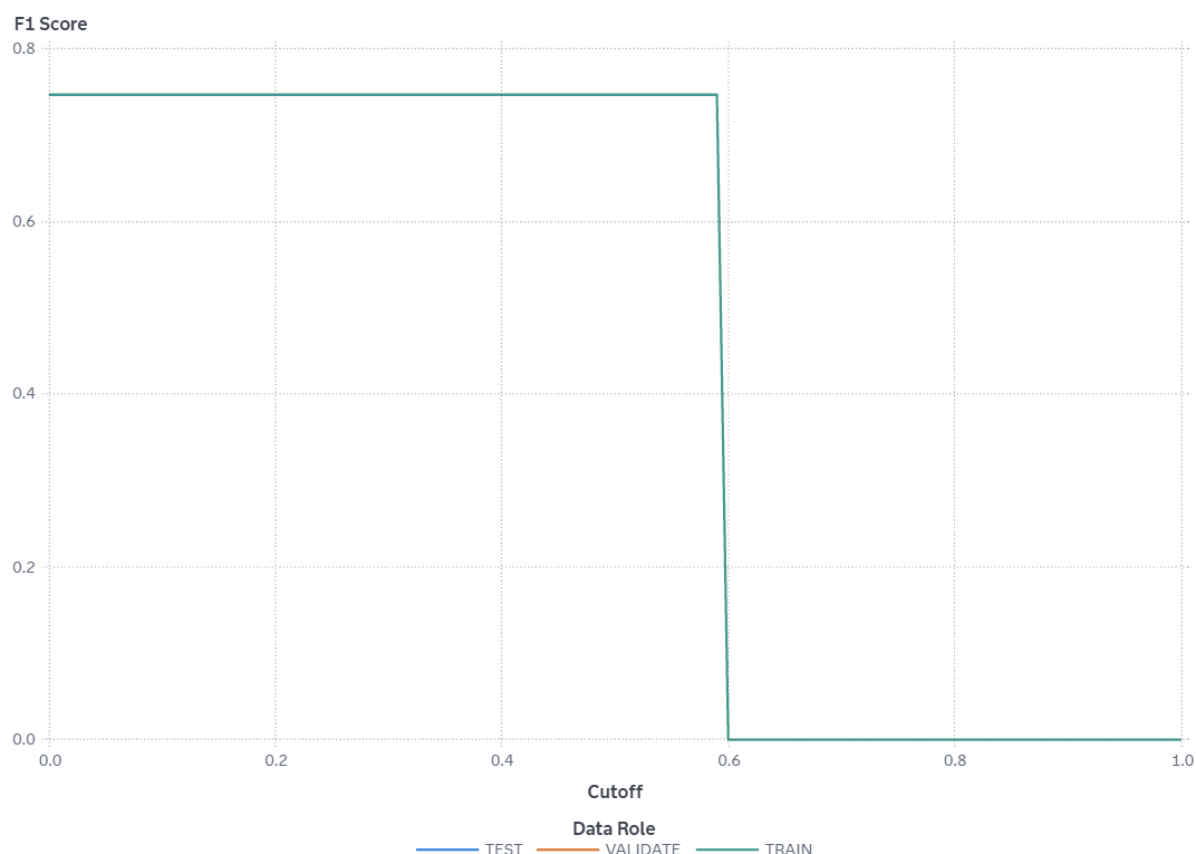
For this model, the accuracy in the TEST partition at the cutoff of 0.5 is 0.596.

For this model, the accuracy in the TRAIN partition at the cutoff of 0.5 is 0.597.

For this model, the accuracy in the VALIDATE partition at the cutoff of 0.5 is 0.597.

Accuracy is the proportion of observations that are correctly classified as either an event or non-event, calculated at various cutoff values. Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether $P_Reached_on_Time_Y_N1$, which is the predicted probability of the event "1" for the target Reached.on.Time_Y.N, is greater than or equal to the cutoff value. When $P_Reached_on_Time_Y_N1$ is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event. When the predicted classification and the actual classification are both events (true positives) or both non-events (true negatives), the observation is correctly classified. If the predicted classification and actual classification disagree, then the observation is incorrectly classified. Accuracy is calculated as $(\text{true positives} + \text{true negatives}) / (\text{total observations})$.

F1 Score



For this model, the F1 score in the TEST partition at the cutoff of 0.5 is 0.747.

For this model, the F1 score in the TRAIN partition at the cutoff of 0.5 is 0.747.

For this model, the F1 score in the VALIDATE partition at the cutoff of 0.5 is 0.747.

The F1 score combines the measures of precision and recall (or sensitivity), which are measures of classification based on the confusion matrix that are calculated at various cutoff values. Cutoff values range from 0 to 1, inclusive, in increments of 0.01. At each cutoff value, the predicted target classification is determined by whether `P_Reached_on_Time_Y_N1`, which is the predicted probability of the event "1" for the target `Reached.on.Time_Y.N`, is greater than or equal to the cutoff value. When `P_Reached_on_Time_Y_N1` is greater than or equal to the cutoff value, then the predicted classification is the event, otherwise it is a non-event.

The confusion matrix for each cutoff value contains four cells that display the true positives for events that are correctly classified (TP), false positives for non-events that are classified as events (FP), false negatives for events that are classified as non-events (FN), and true negatives for non-events that are classified as non-events (TN). True negatives include non-event classifications that specify a different non-event.

Precision is calculated as $TP / (TP + FP)$, and recall (or sensitivity) is calculated as $TP / (TP + FN)$. The F1 score is calculated as $2 * Precision * Recall / (Precision + Recall)$, which is the harmonic mean of Precision and Recall. Larger F1 scores indicate a more accurate model.

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
Reached.on.Time_ Y.N	TEST	2	2
Reached.on.Time_ Y.N	TRAIN	1	1
Reached.on.Time_ Y.N	VALIDATE	0	0

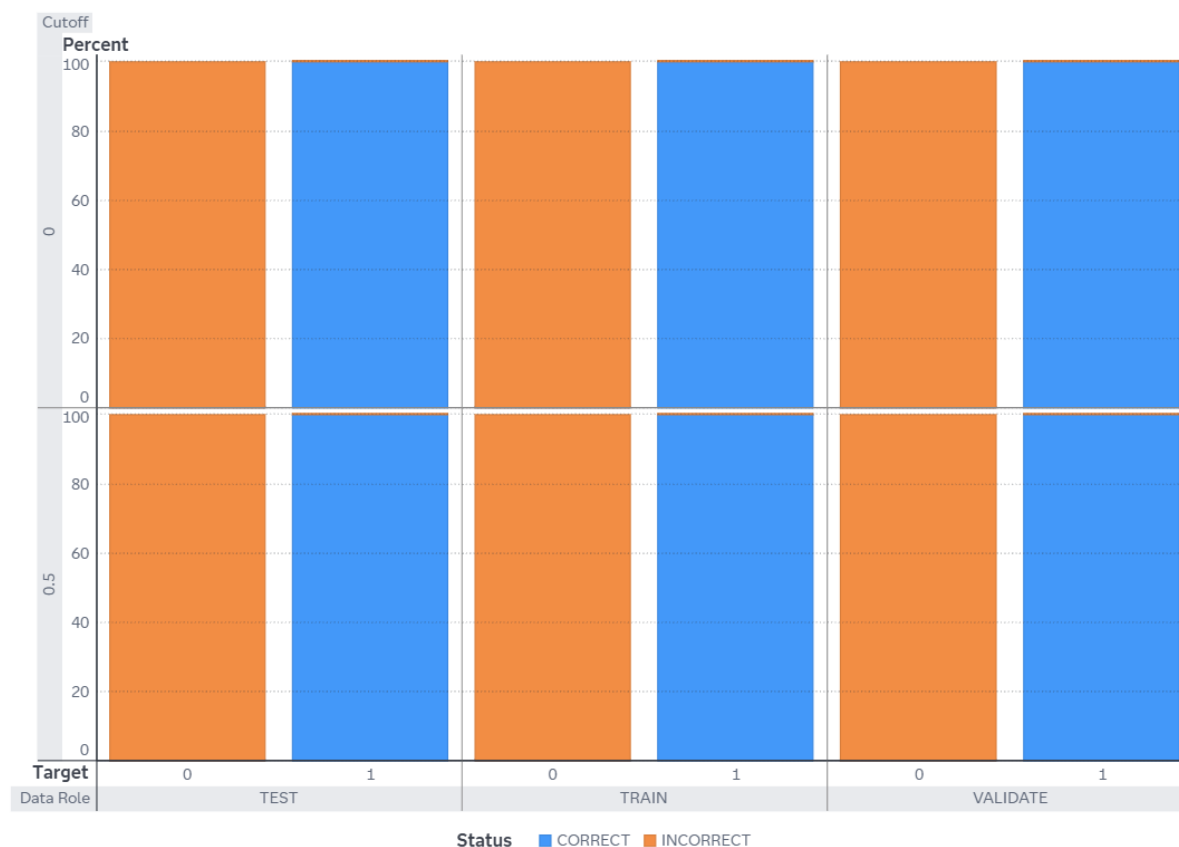
Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,100	0.2407	1,100	0.4906
6,599	0.2406	6,599	0.4905
3,300	0.2407	3,300	0.4906

Misclassification Rate	Multi-Class Log Loss	KS (Youden)	Area Under ROC
0.4036	0.6745	0	0.5000
0.4032	0.6743	0	0.5000
0.4033	0.6743	0	0.5000

Gini Coefficient	Gamma	Tau	KS Cutoff
0		0	0
0		0	0
0		0	0

KS at User-Specified Cutoff	Misclassification Rate at KS Cutoff (Event)	Misclassification Rate (Event)
0	0.4036	0.4036
0	0.4032	0.4032
0	0.4033	0.4033

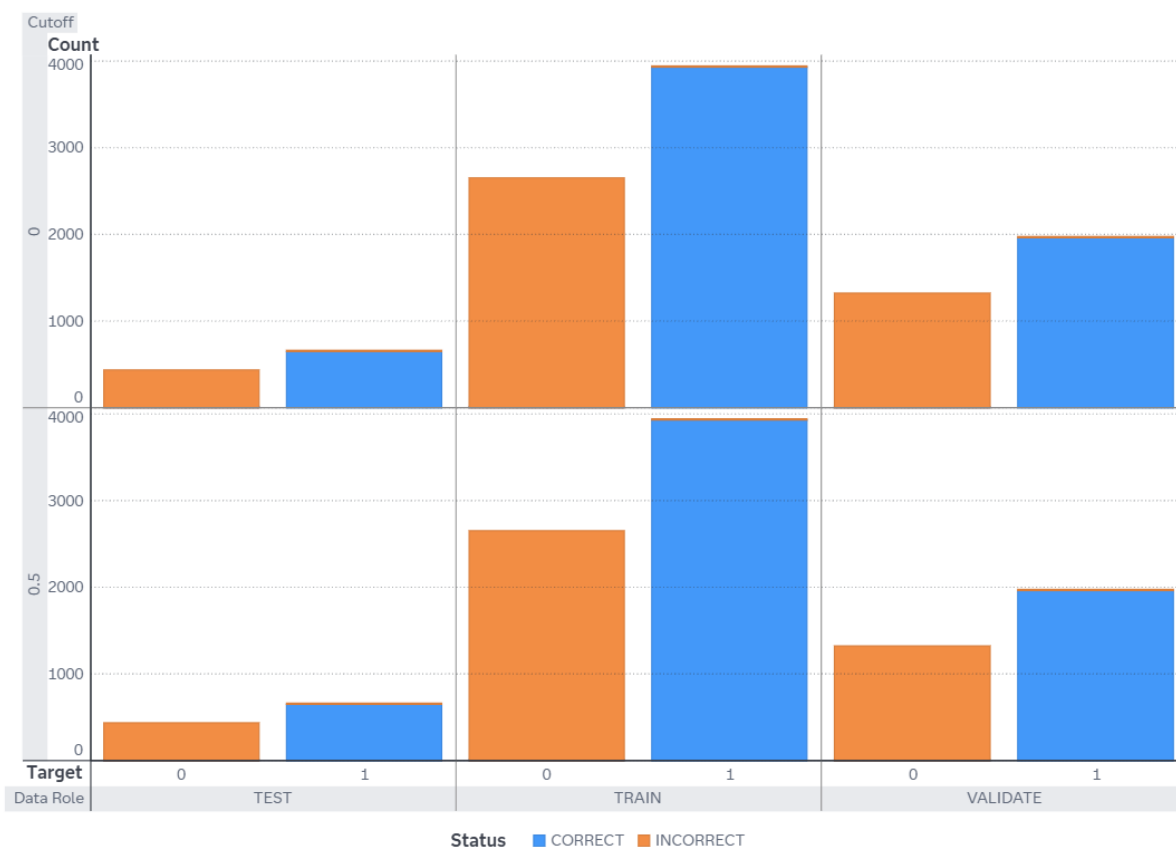
Percentage Plot



The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0 (TRAIN), 0 (VALIDATE), 0 (TEST).

For this data, for the bar corresponding to the event level of Reached.on.Time_Y.N, "1", the segment of the bar colored as "CORRECT" corresponds to true positives.

Count Plot



The Event Classification report is a visual representation of the confusion matrix at various cutoff values for each partition. The classification cutoffs used in the plot are the default (0.5) and these KS cutoff values for existing partitions: 0 (TRAIN), 0 (VALIDATE), 0 (TEST).

For this data, for the bar corresponding to the event level of Reached.on.Time_Y.N, "1", the segment of the bar colored as "CORRECT" corresponds to true positives.

Table

Cutoff	Cutoff Source	Target Name	Response
0	KS	Reached.on.Time_ Y.N	CORRECT
0	KS	Reached.on.Time_ Y.N	INCORRECT
0	KS	Reached.on.Time_ Y.N	CORRECT
0	KS	Reached.on.Time_ Y.N	INCORRECT
0.5000	Default	Reached.on.Time_ Y.N	CORRECT
0.5000	Default	Reached.on.Time_ Y.N	INCORRECT
0.5000	Default	Reached.on.Time_ Y.N	CORRECT
0.5000	Default	Reached.on.Time_ Y.N	INCORRECT

Event	Value	Training Frequency	Validation Frequency
1	True Positive	3,938	1,969
1	False Negative	0	0
0	True Negative	0	0
0	False Positive	2,661	1,331
1	True Positive	3,938	1,969
1	False Negative	0	0
0	True Negative	0	0
0	False Positive	2,661	1,331

Test Frequency	Training Percentage	Validation Percentage	Test Percentage
656	100	100	100
0	0	0	0
0	0	0	0

Test Frequency	Training Percentage	Validation Percentage	Test Percentage
444	100	100	100
656	100	100	100
0	0	0	0
0	0	0	0
444	100	100	100

Properties

Property Name	Property Value
binaryProbCutoff	0.5000
chooseCriterion	SBC
classCoding	GLM
classOrder	FMTASC
codeLocation	mlearning
dataMiningVersion	V2025.08
exactPctlLift	true
explainFidelity	false
explainInfo	false
factorInteractions	false
factorSplit	false
fullDatasetReconstitution	false
hierarchy	NONE
icePlots	false
informativeMiss	false
linkFunction	LOGIT
maxEffects	0
maxNumShapVars	20
maxSteps	0
minEffects	0
missAsLvl	false
nBins	50
nomlinkFunction	GLOGIT
normalize	true
pdNumImportantInputs	5
pdObsSamples	1,000
pdPlots	false
performKernelShap	false

Property Name	Property Value
performLime	false
performVI	false
polynomialDegree	2
seedId	12,345
selectCriterion	SBC
selectMethod	STEPWISE
slEntry	0.0500
slStay	0.0500
specifyRows	RANDOM
stopCriterion	SBC
suppressIntercept	false
tech	NRRIDG
templateRevision	2
train	true
truncateLI	5
truncateUI	95
usePolynomial	false
useSpline	false
useSplineSplit	false
userProbCutoff	false

Output

The SAS System

The GENSELECT Procedure

Model Information	
Data Source	DM_7Y3MJA0HMF8HNCUULOQ0HTKY
Response Variable	Reached.on.Time_Y_N
Distribution	Binary
Link Function	Logit
Optimization Technique	NewtonRaphson with Ridging
Predicted Response Level	I_Reached_on_Time_Y_N

Number of Observations				
Description	Total	Training	Validation	Testing
Number of Observations Read	10999	6599	3300	1100
Number of Observations Used	10999	6599	3300	1100

Response Profile					
Ordered Value	Reached.on.Time_Y_N	Total Frequency	Training	Validation	Testing
1	0	4436	2661	1331	444
2	1	6563	3938	1969	656

Probability modeled is Reached.on.Time_Y_N = 1.

Class Level Information		
Class	Levels	Values
Customer_care_calls	6	2 3 4 5 6 7
Customer_rating	5	1 2 3 4 5
Gender	2	F M
Mode_of_Shipment	3	Flight Road Ship
Prior_purchases	8	2 3 4 5 6 7 8 10
Product_importance	3	high low medium
Warehouse_block	5	A B C D F

Selection Information	
Selection Method	Stepwise
Select Criterion	SBC
Choose Criterion	SBC
Stop Criterion	SBC
Effect Hierarchy Enforced	None
Stop Horizon	3

Selection Details

Convergence criterion (GCONV=1E-8) satisfied

Selection Summary				
Step	Effect Entered	Effect Removed	Number Effects In	SBC
0	Intercept		1	8908.2677
1	Discount_offered		2	7900.5241
2		Discount_offered	1	7894.2982

* Optimal Value Of Criterion

Stepwise selection stopped because adding or removing an effect does not improve the SBC criterion

The model at step 2 is selected where SBC is 7894.298

Selected Effects Intercept

Selected Model

Dimensions	
Columns in Design	1
Number of Effects	1
Max Effect Columns	1
Rank of Design	1
Parameters in Optimization	1

Fit Statistics			
Description	Training	Validation	Testing
-2 Log Likelihood	8899.47299	4450.64471	1483.80955
AIC (smaller is better)	8901.47299	4452.64471	1485.80955
AICC (smaller is better)	8901.47359	4452.64592	1485.81319
SBC (smaller is better)	8908.26766	4458.74639	1490.81262
Average Square Error	0.24064	0.24066	0.24071

Parameter Estimates				
Parameter	DF	Estimate	Standard Error	Chi-Square Pr > ChiSq
Intercept	1	0.391971	0.025095	243.9778 <.0001

Score Code Variables for Predicted Probability	
Reached.on.Time_Y_N	Variable
1	P_Reached_on_Time_Y_N1
0	P_Reached_on_Time_Y_N0

Task Timing		
Task	Seconds	Percent
Setup and Parsing	0.01	11.09%
Levelization	0.01	11.46%
Model Initialization	0.00	1.30%
SSCP Computation	0.02	18.83%
Model Selection	0.06	55.84%
Producing Score Code	0.00	1.12%
Display	0.00	0.26%
Cleanup	0.00	0.00%
Total	0.10	100.00%

